

Overview

1. Intro

Combat is a turn-based battle between two opposing teams.

Each team consists of one or more combat units. The player's team also contains a **hero**. The enemy team may optionally contain a hero as well.

Heroes and regular units share the same battlefield and, once the hero is placed on the grid, follow the same basic positioning and combat rules. Their main difference is how their turns are controlled:

- **Regular units** act automatically according to their combat behavior.
- **Heroes** are controlled manually by the player.

Combat does not alternate between the two teams. Instead, all combatants participate in a shared **initiative system** that determines which unit acts next.

A combat proceeds until one team has no living units remaining.

2. Regular Units and Heroes

Regular Units

Regular units are autonomous combatants. The player does not directly control their actions.

When a regular unit's turn begins, it makes its own decisions based on the current combat state.

Its current priority is:

1. Cast a skill if it has enough mana and can find a valid target.

2. Otherwise, attack an appropriate enemy if one can be targeted.
3. Otherwise, wait.

The unit's behavior is therefore determined by its current:

- Stats
- Position
- Mana
- Available skill
- Skill targeting rules
- Attack range
- Movement range
- Attack targeting rules
- Active effects
- Other combat conditions

Heroes

Heroes participate in the same combat as regular units but are controlled manually.

A hero is intended to occupy a normal hex on the battlefield rather than standing outside the combat grid.

The hero therefore uses the same:

- Positioning system
- Movement system
- Attack system
- Damage system
- Effects system
- Initiative system

The important distinction is that the player decides what the hero does when its turn begins.

This makes the hero less of a separate combat entity and more of a **manually controlled combatant** within the same combat system.

Regular units automatically determine their actions and targets, while the player manually determines the hero's actions and targets.

3. Unit Stats

Combat units currently have several statistics that directly influence their behavior.

Maximum HP

Determines the maximum amount of health a unit can have.

The current HP is derived from the amount of HP the unit has already lost:

$$\text{Current HP} = \text{Maximum HP} - \text{HP Spent}$$

A unit is considered alive while its remaining HP is greater than zero.

Attack Damage

Determines the base physical damage dealt by the unit's normal attack.

Attack Range

Determines how close an enemy must be for the unit to attack it.

Because combat takes place on a hex grid, attack range is measured using hex-grid distance.

Movement Range

Determines how far the unit can move during a turn when trying to reach an enemy.

Required Mana

Determines how much mana the unit needs before it can cast its skill.

Initiative

Determines when the unit gets an opportunity to act relative to other combatants.

Unit stats can also be modified during combat through the stat modification system.

4. Initiative

Combat uses a shared, invisible initiative timeline.

There are no fixed player and enemy turns. Instead, every combatant participates in the same initiative system.

The initiative system determines which unit should act next based on the current initiative state of all combatants.

When initiative advances, the next active unit is selected and its turn begins.

At the beginning of a unit's turn, turn-start effects are processed before the unit performs its actions.

The general flow is:

1. Determine the next unit according to initiative.
2. Make that unit active.
3. Process effects that trigger at the start of its turn.
4. Let the unit perform its turn.

5. Advance initiative.
6. Select the next unit.

This allows units with different initiative characteristics to act at different frequencies instead of simply alternating between the two teams.

The exact mathematical behavior of initiative is not yet documented here because it depends on the implementation of the initiative system itself.

5. Regular Unit Turns

Regular units follow an automatic decision hierarchy.

When a unit becomes active, it first checks whether it can cast its skill.

A unit can cast when:

- It has a skill.
- Its current mana is at least the skill's required mana.
- The skill can find a valid target.

If these conditions are satisfied, the unit casts its skill.

If the unit cannot cast, it attempts to find an enemy to attack.

If it finds a suitable target, it attempts to attack it.

If no attack is available, the unit waits.

The current priority is therefore:

1. **Skill**
2. **Attack**
3. **Wait**

This is the core automatic behavior of regular combat units.

6. Movement

Combat takes place on a hexagonal grid.

Every unit has a current hex position and a movement range.

Movement is currently driven primarily by the unit's attack behavior.

A regular unit does not independently decide that it wants to move somewhere. Instead, it first chooses an enemy to attack.

If the enemy is already within attack range, the unit can attack immediately.

If the enemy is outside attack range, the unit attempts to find a path that brings it into attack range.

The unit can only travel up to its movement range during the current turn.

If the unit cannot reach attack range in one turn, it moves as far as possible and the turn ends. On a later turn, it can continue moving toward the target.

This means movement is currently a consequence of an attack decision:

The unit chooses an enemy → checks whether it can attack → moves toward the enemy if necessary.

This keeps regular-unit behavior relatively simple and predictable.

7. Targeting

Target selection is an important part of combat because different actions can have different targeting rules.

Regular attacks use the unit's attack-targeting rules to determine which enemy should be attacked.

Skills use their own target-selection rules.

Targeting is separate from the effect that ultimately happens to the target. This allows different skills to combine different ways of selecting targets with different effects.

For example, a skill could automatically select:

- The lowest-health ally.
- The nearest enemy.
- A random enemy.
- Multiple enemies.
- A specific area.
- Another unit satisfying some condition.

The targeting system determines **what the skill targets**, while the skill's cast determines **what happens to that target**.

8. Unit Skills

A regular unit skill consists of two main parts:

- **Target selection**
- **Cast**

The target-selection part determines which target the unit should choose.

The cast determines what the skill actually does.

For an automatically controlled unit, target selection is performed automatically.

When a unit decides to cast:

1. The unit checks that it has enough mana.
2. The skill's target-selection rules search the current combat state.
3. A valid target is selected.

4. The skill's cast is performed against that target.
5. The required mana is spent.
6. The skill may either end the unit's turn or allow it to continue, depending on the skill.

This separation allows skills to be constructed from reusable targeting and casting behaviors.

For example:

- A healing cast can be combined with "lowest HP ally" targeting.
 - A damage cast can be combined with "nearest enemy" targeting.
 - A buff cast can be combined with "random ally" targeting.
 - An area-damage cast can be combined with an area-selection rule.
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9. Hero Skills

Hero skills use the same basic cast concept as regular unit skills, but their targeting works differently.

A hero skill consists of:

- **A cast**
- **A targeting mode**

The targeting mode describes what kind of target the skill accepts. The player then manually selects the actual target or target area.

For example, a hero skill could use a targeting mode that requires:

- A single enemy.
- A single ally.
- A specific hex.
- An area around a selected hex.

- A line.
- A cone.
- The hero itself.

The important distinction is that the game does not automatically choose the target for the hero.

Instead:

1. The player selects the hero's skill.
2. The skill's targeting mode determines what the player is allowed to select.
3. The player selects the target or target area.
4. The cast is executed against the selected target.
5. The skill's mana and turn rules are applied.

This means regular and hero skills can share the same underlying cast concepts while using different approaches to target selection.

Regular units use:

Cast + automatic target selection

Heroes use:

Cast + manually selected target constrained by a targeting mode

10. Attacks

A normal attack is the basic physical offensive action available to a unit.

The attack process consists of several stages.

First, the attacker selects an enemy target according to its attack-targeting rules.

The game then checks whether the target is within attack range.

If necessary, the attacker moves toward the target before attacking.

Once the target is within range, the attack is performed.

The attack then goes through the following gameplay steps:

1. Check whether the attack is dodged.
2. If it is not dodged, trigger attack-related effects.
3. Apply physical damage to the target.
4. Grant mana to the attacker.

The basic attack currently uses the attacker's attack-damage stat as its base damage value.

11. Dodge

Attacks can be avoided through the dodge system.

The dodge check happens before the attack deals its normal damage.

When an attack is dodged:

- The target does not receive the normal attack damage.
- The attack does not proceed through its normal attack-effect processing.
- The attacker does not receive the normal mana reward from that attack.

Dodge therefore represents a complete avoidance of the attack rather than simply reducing its damage.

12. Mana

Mana allows units to use their skills.

Each unit tracks how much mana it has accumulated, while its stats determine how much mana is required to cast its skill.

A unit becomes capable of casting once:

| Current Mana $\geq$ Required Mana

When the skill is cast, the required amount of mana is spent.

Regular attacks currently generate mana for the attacking unit.

The current implementation uses a fixed value of **7 mana per successful basic attack**. This should be considered a temporary value rather than a finalized combat rule.

The intended gameplay loop is:

1. Unit attacks.
2. Unit gains mana.
3. Unit eventually reaches its required mana.
4. Unit casts its skill.
5. Mana is spent.
6. Unit begins building mana again.

13. Damage and Death

Damage reduces a unit's remaining HP.

When a unit's remaining HP reaches zero or below, it dies.

A dead unit can no longer participate normally in combat.

If the unit that dies is currently the active unit, the initiative system immediately moves on to the next combatant rather than allowing the dead unit to continue its turn.

Death therefore has an immediate effect on the initiative flow.

14. Effects and Status Effects

Combat supports persistent effects that can modify units or react to combat events.

Effects can have:

- Charges.
- Durations.
- Trigger conditions.
- Effects that occur at specific moments in combat.

The currently established trigger points include **turn start** and **attack**.

Turn-start effects

Turn-start effects activate when a unit's turn begins.

This can be used for mechanics such as:

- Damage over time.
- Regeneration.
- Temporary buffs.
- Temporary debuffs.
- Other effects that happen once per turn.

After the turn-start effects are processed, turn-based effect durations are consumed.

Attack effects

Attack effects activate when a unit performs an attack.

This provides a way to implement mechanics that react to or modify attacks, such as:

- On-hit effects.
- Additional damage.
- Temporary attack bonuses.
- Effects applied when attacking.

Effects can have durations measured in different units of combat activity, including turns and attacks.

This allows status effects to be expressed in terms of combat events rather than real time.

15. Combat End Condition

Combat ends when one team has no living units remaining.

At the relevant points in combat, the game checks which teams still have at least one living unit.

Combat continues while both teams have living units.

Once only one team remains alive, the battle is finished.

The combat therefore follows a simple overall loop:

1. Determine the next combatant using initiative.
2. Start that combatant's turn.
3. Process turn-start effects.
4. Let the combatant perform its actions.
5. Apply damage, deaths, effects, movement, and other consequences.
6. Check whether one team has been eliminated.
7. If combat is still ongoing, advance initiative.

8. Repeat.

16. Overall Combat Model

The combat system combines an **auto-chess style unit system** with a **manually controlled hero**.

Regular units are autonomous. Their turns are driven by a simple decision hierarchy:

Cast if possible → otherwise attack → otherwise wait.

Their targeting, movement, and actions are determined automatically from their combat rules and the current state of the battlefield.

The hero exists within the same battlefield and uses the same fundamental combat mechanics, but its decisions are made by the player.

The hero therefore provides the player's direct influence over an otherwise autonomous battle.

The overall design can be summarized as:

- **Two opposing teams** fight on a shared hex grid.
- **Regular units** act automatically.
- **Heroes** act manually.
- **Initiative** determines which combatant acts next.
- **Movement** allows units to approach enemies when they cannot attack from their current position.
- **Targeting** determines who or what an action affects.
- **Unit skills** combine automatic target selection with a cast.
- **Hero skills** combine a cast with a manually selected target constrained by a targeting mode.

- **Attacks** deal physical damage, can be dodged, trigger attack effects, and generate mana.
- **Mana** allows units to use their skills.
- **Effects** provide persistent buffs, debuffs, and event-triggered mechanics.
- **Death** removes units from combat and can interrupt the current initiative turn.
- **Combat ends** when one team has no living units.